Per-site retrieval of regolith thermal conductivity at the Apollo 15 and 17 heat-flow boreholes

Models, method, and why the result holds — a walkthrough

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Purpose. This note lays out, end to end, what we did and why it is defensible: how the two published conductivity models are built and why the literature uses a single “global” K_d; how we instead retrieve K_d *separately* at each borehole; what changed between the old and new steady-state solver and why the two give the same answer; a physical reading of why the two sites differ; and the role of the basal heat flux Q_b. (The statistical machinery — bootstrap intervals, model selection — is deferred; here the focus is the physics and the method.)

1 The goal and the two sites

The deep regolith thermal conductivity K_d sets how steeply temperature rises with depth and how much of the Moon’s interior heat reaches the surface. The Apollo Heat-Flow Experiment (HFE) left buried temperature sensors at two very different places (Fig. 1): **Apollo 15** at Hadley Rille on the Imbrium mare/Apennine boundary, and **Apollo 17** in the Taurus–Littrow valley among highland massifs and dark pyroclastic mantle. The published thermal models use *one* global K_d for the whole Moon. Our question is simple: *what is K_d at each of these boreholes, separately?*

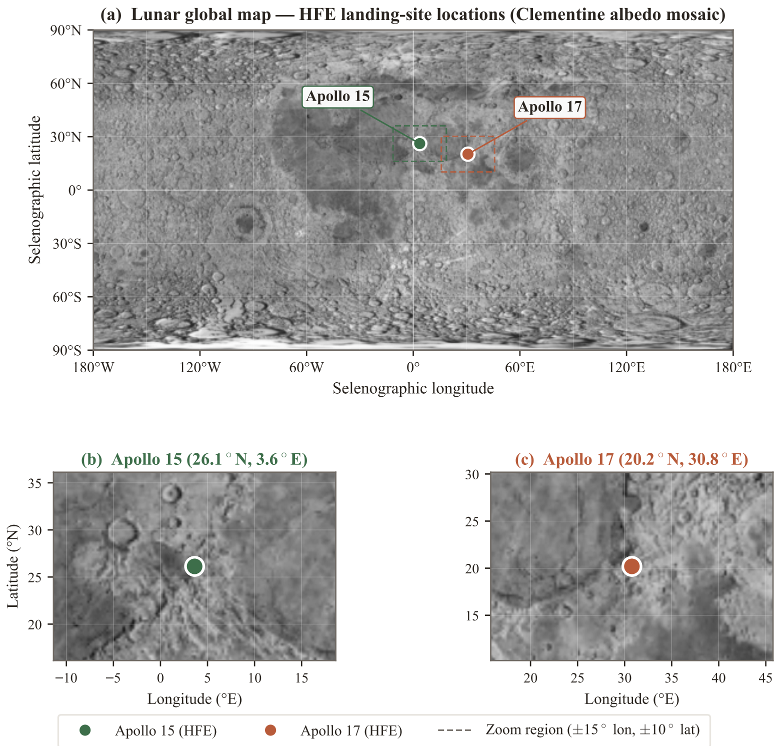

Figure 1: The two heat-flow boreholes sit in geologically distinct terrains — a first hint that a single global conductivity may not fit both.

2 The two published conductivity models (and how to reproduce them)

Both models give the regolith conductivity K as a function of temperature T and depth z. Reproducing them is just evaluating the formulas below; our code does exactly this in `lunar/properties.py`.

2.1 Hayne et al. (2017): the H-parameter model

The regolith is loose and porous at the surface and compacts with depth, so the solid (contact) conductivity rises from a surface value K_s to a deep value K_d over an e-folding depth H:

$$K_c(z) = K_d - (K_d - K_s)\, e^{-z/H}. \tag{1}$$

Radiation across pore spaces then adds a strongly temperature-dependent term:

$$K(T, z) = K_c(z) \left[1 + \chi\, (T/T_{\mathrm{ref}})^3 \right], \qquad T_{\mathrm{ref}} = 350\,\mathrm{K}. \tag{2}$$

The bulk density follows the same compaction law, $\rho(z) = \rho_d - (\rho_d - \rho_s)e^{-z/H}$. The cited parameters are:

Symbol	Value	Meaning
K_s	$0.74\ \mathrm{mW\,m^{-1}\,K^{-1}}$	surface contact conductivity
K_d	$3.4\ \mathrm{mW\,m^{-1}\,K^{-1}}$ **(global)**	**deep contact conductivity — the value we refine**
H	$0.06\,\mathrm{m}$	compaction e-folding depth
χ	2.7	radiative-term strength
T_{ref}	350 K	radiative normalisation
ρ_s, ρ_d	$1100,\ 1800\ \mathrm{kg\,m^{-3}}$	surface / deep bulk density

Why the literature value is $K_d = 3.4$. Hayne et al. fit Eqs. (1)–(2) to the *globally averaged* diurnal surface-temperature curves measured from orbit by the Diviner radiometer. A single (K_s, K_d, H, χ) best reproduced the global day–night behaviour, and $K_d = 3.4\ \mathrm{mW\,m^{-1}\,K^{-1}}$ is that global-average deep asymptote. It was never meant to be site-specific — which is exactly the gap we fill. Fig. 2(a) shows where that asymptote sits and where our per-site values move it.

2.2 Martínez & Siegler (2021): a density-based model

This model writes the conductivity directly in terms of bulk density ρ, splitting it into a contact (phonon) term and a radiative term:

$$K(T, \rho) = (A_1\rho + A_2)\, k_{\mathrm{am}}(T) \; + \; (B_1\rho + B_2)\, T^3, \tag{3}$$

where $k_{\mathrm{am}}(T)$ is the Woods–Robinson et al. (2019) amorphous-solid polynomial and $A_{1,2}, B_{1,2}$ are fitted constants. The two models use different physics for the same quantity; Fig. 2(b) shows they predict comparable total conductivity and the same steep rise with temperature.

3 Our method: retrieving K_d per site

3.1 The forward model

We solve the 1-D heat equation in the regolith column,

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(K(T, z)\, \frac{\partial T}{\partial z} \right), \tag{4}$$

on a grid that is millimetre-fine at the top (to resolve the daily thermal wave) and reaches 5 m. The surface obeys a non-linear radiative balance, $(1 - A)\, S(t) = \varepsilon\sigma T_s^4 + K\, \partial_z T|_0$, and the bottom admits the geothermal flux, $K\, \partial_z T = Q_b$. Driven by the periodic Sun, the column settles to a periodic steady state; that settled profile is what we compare to the sensors.

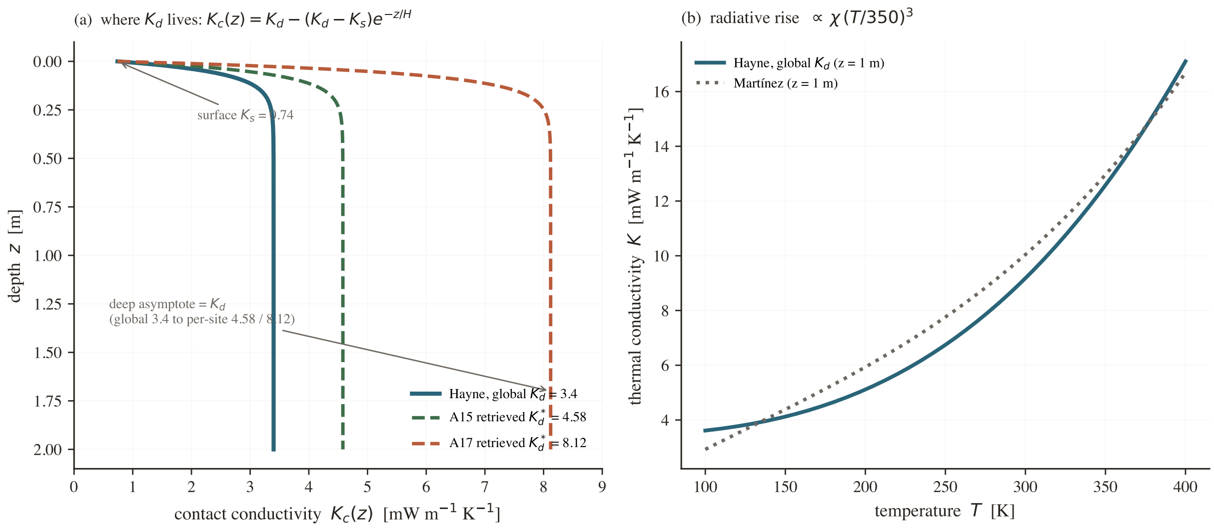

Figure 2: **Reproducing both models.** (a) Hayne contact conductivity $K_c(z)$, Eq. (1): it climbs from K_s at the surface to its deep asymptote K_d. The published *global* value is 3.4; our per-site retrievals push it to 4.58 (A15) and 8.12 (A17). (b) Full conductivity vs. temperature at 1 m, Eq. (2), for Hayne and Martínez — the radiative $\propto T^3$ term roughly doubles K by lunar midday. Both panels are evaluated directly from the model code.

3.2 The retrieval

We hold the Hayne functional form fixed and treat K_d as the single unknown. For each trial K_d we solve Eq. (4) to steady state and measure the misfit to the deep sensors,

$$\mathrm{RMSE}(K_d) = \sqrt{\tfrac{1}{N}\sum_i \big[\,\langle T\rangle(z_i; K_d) - T_i^{\mathrm{obs}}\,\big]^2}, \qquad K_d^* = \arg\min_{K_d}\ \mathrm{RMSE}(K_d). \tag{5}$$

The misfit curve has a clean single minimum at each site (Fig. 3), and that minimum is the retrieved K_d^*.

3.3 Result

The retrieval gives

$$K^*_{d,\mathrm{A15}} = 4.58\ \mathrm{mW\,m^{-1}\,K^{-1}}, \qquad K^*_{d,\mathrm{A17}} = 8.12\ \mathrm{mW\,m^{-1}\,K^{-1}},$$

i.e. Apollo 17 is about twice as conductive as Apollo 15. With these per-site values the modelled deep temperatures track the data far better than the single global K_d does (Fig. 4); at Apollo 17 the misfit is roughly halved. (Confidence intervals and significance are quantified separately by bootstrap — deferred here.)

4 Old vs. new: how we reach the steady state

Everything above rests on "solve to steady state." That step is where the old code went wrong, and where the new method earns its trust.

4.1 What "steady state" requires

The periodic steady state is pinned by two facts: the surface skin repeats each lunation, and — below the skin, with no internal heat sources — the same flux passes through every layer and equals the bottom flux Q_b. That fixes the deep gradient:

$$K(\langle T\rangle, z)\,\frac{d\langle T\rangle}{dz} = Q_b \quad\Longrightarrow\quad \frac{d\langle T\rangle}{dz} = \frac{Q_b}{K(\langle T\rangle, z)}. \tag{6}$$

The deep profile is therefore *not* free; energy conservation fixes its slope once the temperature at one depth is known.

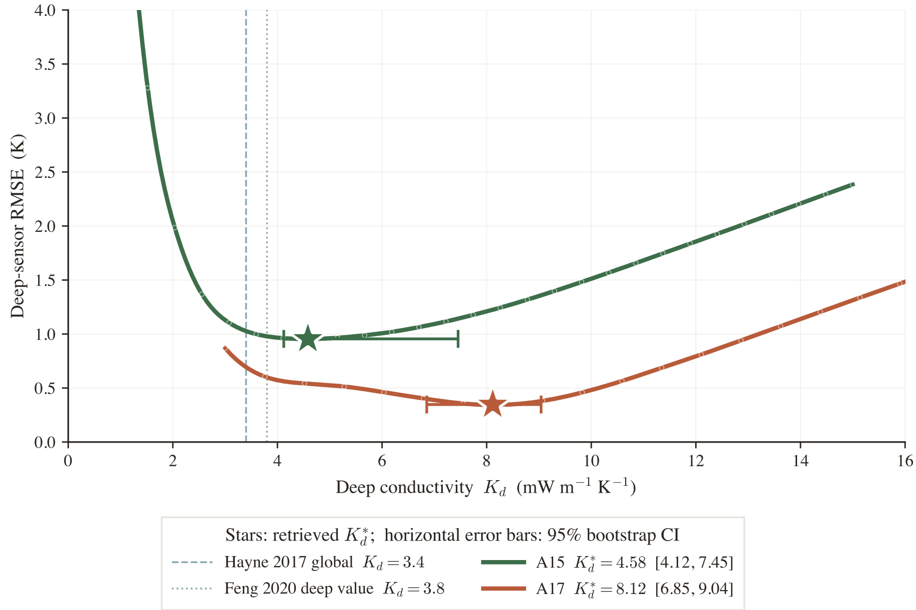

Figure 3: The retrieval, visualised: deep-sensor RMSE vs. trial K_d. Each site has a well-defined minimum, and Apollo 17's sits at a clearly higher conductivity than Apollo 15's.

4.2 The old method, and its hidden bias (flag F1)

The old approach integrated Eq. (4) forward for a fixed, short spin-up (~30 lunations). But the full column relaxes on a diffusive timescale of order 10^3 lunations, so at sensor depth the profile still *remembered the temperature it was started from*. That starting guess then leaked into K_d as a hidden parameter. The trap: the usual "did $\langle T \rangle$ change by < 0.01 K between lunations?" check passes *long before* the column is settled, so the answer looks converged while it is still drifting.

4.3 The new method: a flux-anchored shortcut

The new solver (`lunar/equilibrium.py`) does not wait for diffusion to discover Eq. (6) — it imposes it: (i) a *short* spin-up settles the fast skin and fixes the cycle-mean "anchor" temperature just below it; (ii) from the anchor it integrates Eq. (6) straight down to fill in the whole deep profile; (iii) iterate 3–5 times. It is like computing a tank's steady level from "inflow = outflow" instead of waiting for the sloshing to stop.

4.4 Do they agree? Yes — run the lunations and watch

The cleanest demonstration is to take the *old* method and simply give it more lunations: as the spin-up lengthens, its profile marches monotonically onto the new method's answer (Fig. 5). Starting from two very different temperatures, 240 K and 260 K, both collapse onto the same curve (Fig. 6). They must: both enforce the same two conditions, and the steady state satisfying them is unique — the shortcut just gets there in ~9 s instead of the ~3000 lunations (~4 min) brute force needs.

Takeaway

The new method is not a different model — it reaches the *same* periodic steady state, certified by two diagnostics it returns on every solve (guess-independence ≤ 0.03 K, and the deep flux closing on Q_b to $< 2\%$; Fig. 7), and confirmed by brute force converging onto it. It removes the F1 bias not by running longer but by not needing to.

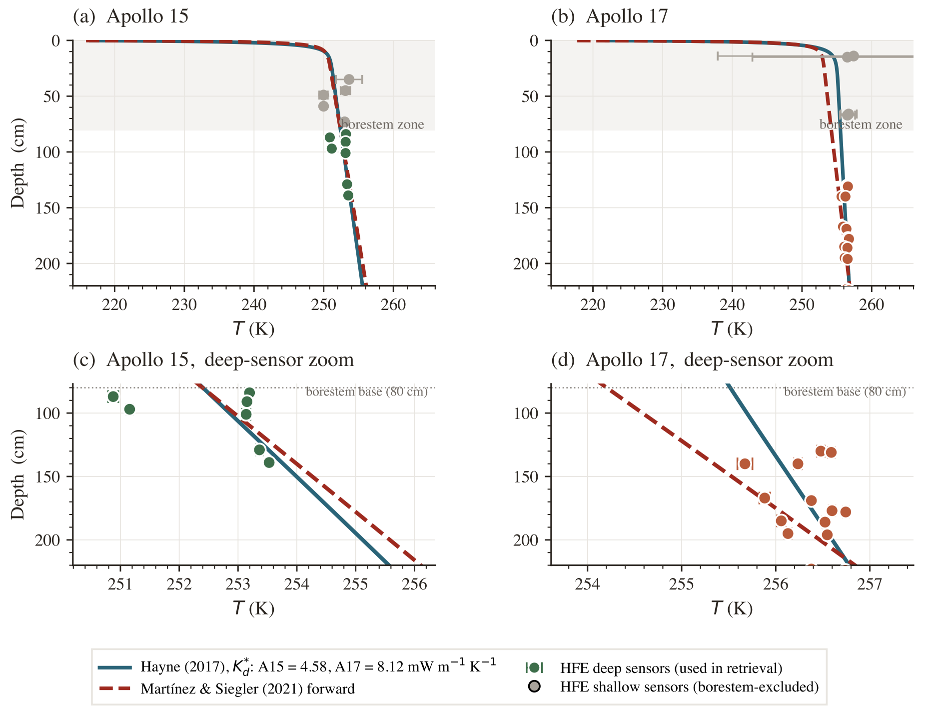

Figure 4: Settled mean temperature vs. depth using each site's *retrieved* K_d^*, against the deep-sensor measurements. The per-site fit threads the data better than the global value can.

5 Why might the two sites differ?

A factor-of-two conductivity contrast is large but physically reasonable: the two boreholes sample different regolith.

	Apollo 15	**Apollo 17**
Setting	Hadley Rille (mare/front)	Taurus–Littrow valley
Latitude	26.1°N	20.2°N
Bond albedo	0.131	0.137
Deepest sensor	1.39 m	2.34 m
Basal flux Q_b (held fixed)	21 mW m^{-2}	15 mW m^{-2}
Retrieved K_d^*	**4.58**	**8.12** mW m^{-1} K^{-1}

Heat moves through dry regolith mainly by grain-to-grain contact, so a higher K_d points to a denser, more compacted, or more rock-rich deep regolith. Plausible contributors at Apollo 17:

- **Material and compaction.** Taurus–Littrow mixes highland massif debris with dense pyroclastic (dark-mantle) material; Hadley samples mare-basalt regolith. Different parent material and packing change grain contact and bulk density.
- **Independent support.** Re-running the retrieval with the *density*-based Martínez model (Eq. (3)) prefers a higher effective deep density at Apollo 17 — the same contrast seen through a model whose free lever is density rather than K_d. Two independent descriptions point the same way.
- **Rock abundance / gardening.** More buried fragments, or a more mature and sintered column, would also raise the effective conductivity.

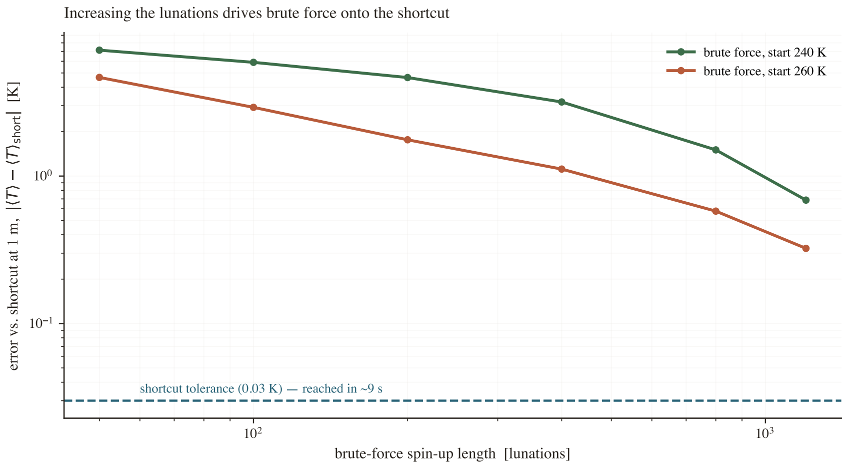

Figure 5: **The "increasing-lunations" graph.** Brute-force error (vs. the shortcut, at 1 m) falls steadily as the spin-up lengthens, heading for the shortcut's certified tolerance (dashed). Two different starting temperatures converge to the *same* place — evidence that the two methods share one solution.

Be honest about

We hold Q_b fixed (next section), so part of the inferred contrast is, in principle, exchangeable with a flux difference. A formal model comparison (deferred) favours a *conductivity* contrast over a flux contrast, but the honest framing is: *Apollo 17's regolith conducts heat about twice as efficiently as Apollo 15's, conditional on the published basal fluxes.*

6 The role of the basal heat flux Q_b

K_d cannot be read off in isolation: the deep gradient is Q_b/K (Eq. (6)), so K_d and Q_b *trade off*. Assume more heat from below and you would infer a higher conductivity to reproduce the same temperatures. The two are entangled, and this single record cannot separate them.

We therefore fix Q_b at the published Apollo values — 21 mW m^{-2} (A15) and 15 mW m^{-2} (A17) — and report K_d *conditional* on them. Fig. 8 maps the trade-off: many K_d–Q_b combinations along a diagonal fit almost equally well, so the absolute K_d shifts if Q_b does. Reassuringly, the *direction* of the inter-site contrast (A17 > A15) persists even when Q_b is allowed to vary, and Q_b is one of the largest single terms in the error budget — a direct fingerprint of this coupling.

7 Why this is defensible — the short version

1. **The models are reproduced from first formulas** (Eqs. (1)–(3), Fig. 2); the global $K_d = 3.4$ is a deliberate global average we are refining, not a measured site value.

2. **The retrieval is transparent**: one unknown, a single RMSE minimum per site (Fig. 3), and a visibly better fit to the data (Fig. 4).

3. **The engine is trustworthy**: the new solver is certified (Fig. 7) *and* independently confirmed by brute force converging onto it as the lunations increase (Fig. 5, 6) — the old hidden-spin-up bias is gone.

4. **The contrast is physically sensible** (denser/different regolith at Apollo 17) and is echoed by an independent density-based model.

5. **We are explicit about Q_b**: K_d is conditional on the assumed basal flux, the trade-off is mapped, and the inter-site *direction* is the robust statement.

Reproduce any of this: `notebooks/02_retrieval.ipynb` (the sweep + result), `notebooks/equilibrium_demo.ipynb` (Fig. 5, 6), and `pytest tests/test_equilibrium.py` (the certification checks).

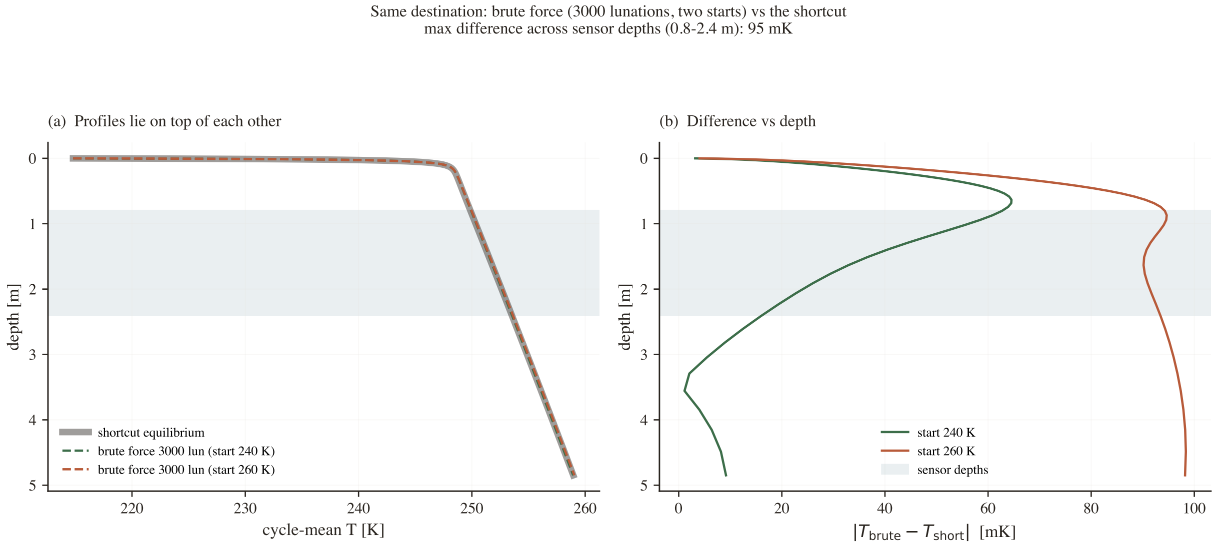

Figure 6: Fully converged brute force from 240 K *and* 260 K (dashed) overlaid on the shortcut (solid). The profiles are indistinguishable; the residual (right, tens of mK) is just brute force finishing its slow descent.

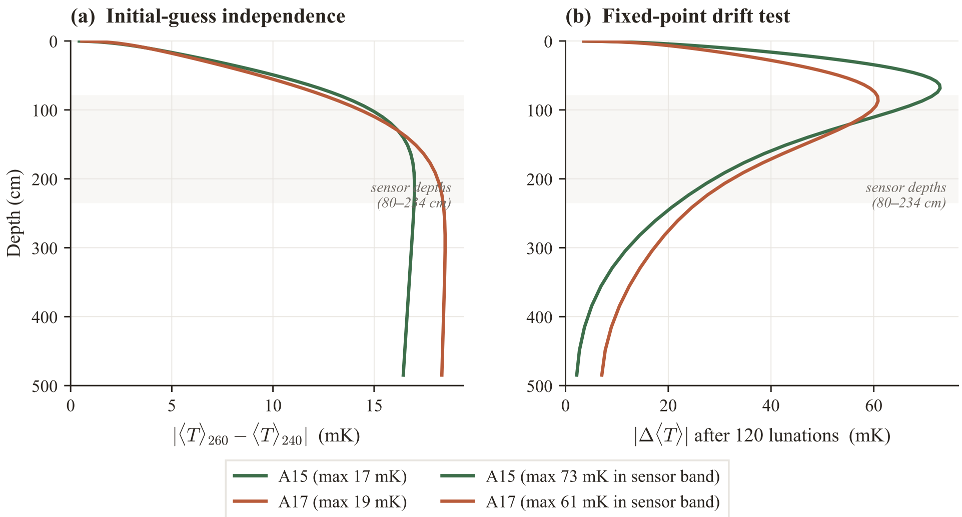

Figure 7: Self-certification: guess-independence and mean-flux closure — the numbers that prove the shortcut arrived at the true steady state.

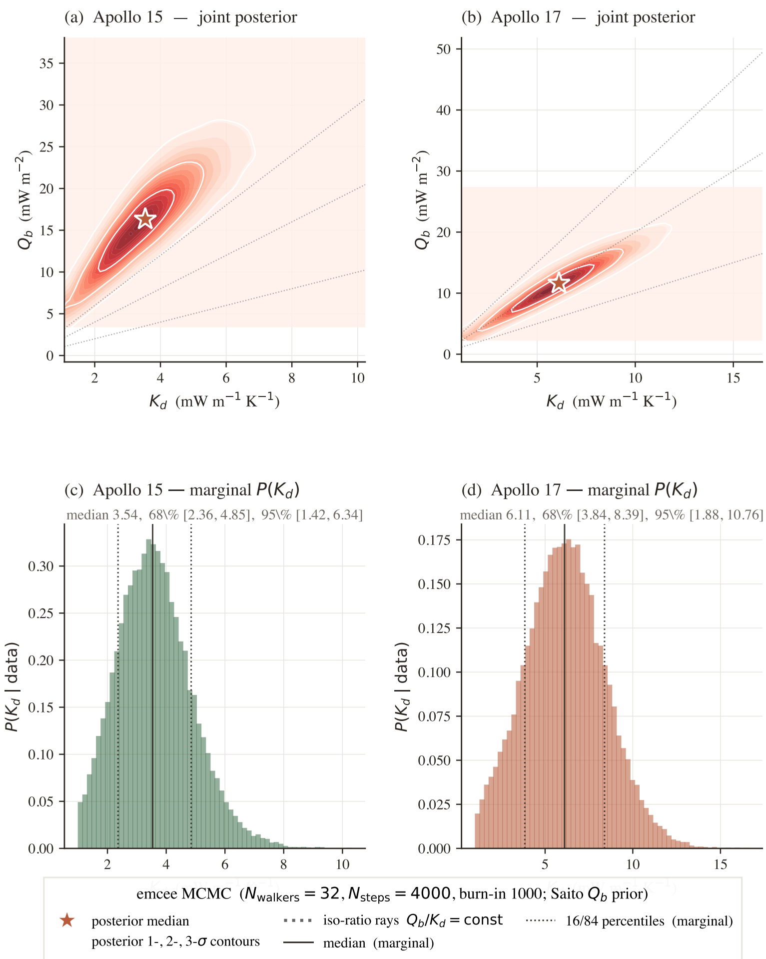

Figure 8: The K_d–Q_b trade-off. Because the deep gradient is Q_b/K, the two are degenerate along a diagonal; the retrieved K_d is conditional on the assumed Q_b. The contrast *between* sites is more robust than either absolute value.